HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and maintained **Java Spring Boot** REST APIs for cross-service data exchange, implementing request validation and circuit-breaker patterns to enhance system resilience.
- Engineered **Python** ETL pipelines for data ingestion and transformation, processing approximately **1.5 TB** of raw data weekly into a structured format for downstream analytics.
- Managed production support for enterprise applications using **Splunk** and **Dynatrace**, resolving P1/P2 incidents within strict SLAs and performing root cause analysis on system failures.
- Collaborated with cross-functional teams, including Quality and Manufacturing Engineering, to identify and implement improvements for product quality and test system efficiency, supporting the resolution of manufacturing issues.
- Performed device failure analysis on implantable electronic medical devices to identify root causes at the component level, contributing to a **15%** reduction in recurring component-level failures.
- Applied data analysis and technical problem-solving skills using internal component and device tracking systems to support manufacturing operations and continuous improvement initiatives.
- Containerized microservices using **Docker** and deployed them to an **AWS EC2** environment, configuring basic resource limits and networking for predictable rollouts.

SKILLS

Programming Languages – Java, Python, SQL, R, C++, C

Frameworks & Libraries – Spring Boot, pandas, NumPy, scikit-learn, Hibernate, JPA, FastAPI, Flask

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git, Linux

Databases – PostgreSQL, MySQL, Oracle

BI & Analytics Tools – Power BI, Tableau, DAX, Power Query, RStudio

Methodologies – Agile, Scrum

Monitoring & Observability – Splunk, Dynatrace, CloudWatch

PROJECTS

Automated Test Fixture Data Analysis System (HCL Technologies)

- Developed a **Python** application to ingest and analyze test data from multiple automated test fixtures, identifying performance bottlenecks and reducing test cycle times by **20%**.
- Implemented data validation routines using **pandas** and **SQL** against a **PostgreSQL** database, ensuring data integrity for over **50,000** test records monthly.
- Created interactive **Power BI** dashboards to visualize test yield trends, failure modes, and key performance indicators for manufacturing engineering stakeholders.

Medical Device Manufacturing Process Improvement (MSc Project)

- Designed and executed test protocols for a new manufacturing process for implantable electronic components, analyzing results to ensure adherence to quality standards.
- Performed root cause analysis on a subset of component failures, identifying a critical soldering defect that was subsequently addressed by the manufacturing team.
- Documented findings and recommendations in a technical report, including statistical analysis performed using **R**, supporting regulatory submission processes.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)